PRODUCT REQUIREMENTS SPECIFICATION & SPRINT BACKLOG
AayurLink

DOCUMENT METADATA

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1. EPIC BREAKDOWN (HIGH-LEVEL MODULES)

- EPIC-01: User Onboarding & Multilingual Access (Auth & Profiles)
- EPIC-02: Dialysis Scheduling & Slot Reservation Engine (Booking System)
- EPIC-03: Longitudinal Biomarker & Health Log Tracker (Vitals Logger)
- EPIC-04: Community Clean Drinking Water Directory (RO Plant Locator)
- EPIC-05: Clinical Admin & Nursing Triage Dashboard (Hospital Portal)

2. DETAILED USER STORIES & ACCEPTANCE CRITERIA (INVEST & GHERKIN STANDARD)

EPIC 01: USER ONBOARDING & ACCESS

[US-01: Low-Friction Mobile Authentication]
As a rural caregiver (Kasun),
I want to log in using my 10-digit mobile phone number,
So that I can quickly access the platform without memorizing complex passwords.

Acceptance Criteria (Gherkin Format):
Scenario: Successful login with mobile number
  Given the user is on the landing page
  When the user selects their preferred language (Sinhala / English)
  And enters a valid 10-digit Sri Lankan phone number (e.g., 077XXXXXXX)
  And inputs the one-time verification code (OTP / test pin)
  Then the user should be redirected to the Patient Home Dashboard.

Scenario: Invalid phone number entry
  Given the user enters a number with fewer or more than 10 digits
  When the user taps "Continue"
  Then the system should display an inline error message in the selected language:
    "කරුණාකර වලංගු දුරකථන අංකයක් ඇතුළත් කරන්න / Please enter a valid mobile number."

EPIC 02: DIALYSIS SCHEDULING ENGINE

[US-02: Dialysis Slot Reservation]
As a caregiver managing patient appointments,
I want to select an available morning or afternoon shift at the regional hospital,

So that my father can undergo hemodialysis without waiting in physical queues.

Acceptance Criteria:
Scenario: Booking an available shift
  Given the patient has no active booking for the chosen date
  When the user selects the Hospital, Date, and "Morning Shift (08:00 - 12:00)"
  And machine capacity is greater than 0
  Then the system reserves 1 machine slot
  And transitions status to "Confirmed"
  And displays a digital booking pass with a unique Token ID (e.g., #SLOT-AN-042).

Scenario: Machine capacity exhausted
  Given all machine slots for the selected shift are booked
  When the user selects that shift
  Then the shift badge should display "Full (පිරී ඇත)"
  And the system should recommend the next available date/shift automatically.

EPIC 03: VITAL HEALTH & LAB TRACKER

[US-03: Serum Creatinine & eGFR Log Entry]
As a caregiver or patient,
I want to log new Serum Creatinine and eGFR values from our clinic report slips,
So that I can see whether the kidney condition is stable over time.

Acceptance Criteria:
Scenario: Successful biomarker entry
  Given the user is on the "My Health Logs" tab
  When the user inputs:
    - Test Date
    - Serum Creatinine value (e.g., 4.2 mg/dL)
    - eGFR value (e.g., 18 mL/min/1.73m²)
  And taps "Save Record"
  Then the system validates numeric boundaries (Creatinine: 0.1 - 25.0)
  And updates the chronological line chart instantly
  And displays status badge (e.g., "Critical / High" based on standard nephrology ranges).

EPIC 04: COMMUNITY CLEAN WATER DIRECTORY

[US-04: Finding Operational RO Plants]
As a CKD patient family member,
I want to filter and view working community RO water stations by district/GN division,
So that we can collect verified safe drinking water without travelling to non-functional units.

Acceptance Criteria:
Scenario: Filter RO plants by region
  Given the user opens the "Clean Water (ජලවර)" tab
  When the user selects "Anuradhapura" -> "Medawachchiya"

Then the system displays a card list and map pins of all RO plants in that area
And shows an indicator badge: "Operational (ක්‍රියාකාරී)" or "Under Maintenance (අක්‍රියයි)".

EPIC 05: CLINICAL ADMIN PORTAL

[US-05: One-Tap Nurse Check-In Roster]
As a clinic nurse (Sujatha),
I want to view today's scheduled dialysis roster and tap "Check-in" when a patient arrives,
So that bed occupancy is tracked without writing in paper logbooks.

Acceptance Criteria:
Scenario: Checking in an arriving patient
Given Nurse Sujatha is logged into the Hospital Admin Dashboard
When she views today’s shift roster (e.g., Morning Shift: 10 Beds)
And patient Kasun presents Token #SLOT-AN-042
And she taps the "Check-in" button on that record
Then the bed status changes from "Reserved" to "Occupied (In-Progress)"
And the timestamp is logged automatically.